

Last-Minute Revision Sprint (90 to 180 Minutes)

Use this right before interview day.

1) 90-Minute High-Impact Plan

0-15 min:

- Review Go concurrency pitfalls and context cancellation.
- Rehearse channel vs mutex, pointer vs value receiver, nil interface pitfall.

15-30 min:

- Review AWS architecture choices: ECS/EKS/Lambda, RDS/Aurora/DynamoDB, SQS/SNS/EventBridge/MSK.
- Rehearse one migration and one incident story.

30-45 min:

- Review Kubernetes probes, autoscaling, rollout safety, and common debugging flow.

45-60 min:

- Review Postgres indexes, explain analyze, transactions, locking, replication lag discussion.

60-75 min:

- Review Kafka reliability semantics, lag handling, idempotency, and schema compatibility.

75-90 min:

- Review Redis cache patterns and RabbitMQ retry/DLQ routing.
- End with 5 rapid-fire design questions.

2) Extended 180-Minute Plan

- Spend extra 30 min on Java backend talking points.
- Spend extra 30 min on observability and SLO-based incident response.
- Spend extra 30 min rehearsing 8 STAR stories with numbers.

3) 20 Must-Know Go Q and A

1. Why use context everywhere?
 - cancellation, deadlines, and request-scoped propagation.
2. Channel vs mutex?
 - channels for coordination/ownership transfer, mutex for shared mutable state.
3. Why can map crash in concurrent writes?
 - native map is not write-safe concurrently.
4. What causes goroutine leaks?
 - blocked channel ops and missing cancellation paths.
5. How to profile CPU and memory quickly?
 - pprof with representative traffic.
6. Why preallocate slices?

- fewer reallocations and copy overhead.
7. How to avoid nil interface bug?
 - return nil explicitly and use typed errors carefully.
 8. How do you propagate timeouts to DB and HTTP?
 - context.WithTimeout and context-aware client APIs.
 9. How do you do safe retries?
 - transient failures only + jitter + idempotency.
 10. How do you do graceful shutdown?
 - stop intake, drain requests, cancel workers, close resources.
 11. What is outbox pattern?
 - atomic DB write + deferred reliable publish.
 12. How to ensure idempotency for create APIs?
 - idempotency key with unique constraints.
 13. Why race detector is useful?
 - catches data races not obvious in tests.
 14. Why can default case in select be risky?
 - can cause busy loop and high CPU.
 15. Why can time.After in loop be problematic?
 - timer allocation/leak style overhead; prefer timer reuse.
 16. How do you structure packages in backend services?
 - cohesive boundaries and explicit dependencies.
 17. What is a good error taxonomy?
 - validation/not found/conflict/transient/internal.
 18. How do you reduce lock contention?
 - shorten critical section, sharding, lock-free paths where possible.
 19. How do you discuss optimization in interviews?
 - baseline, bottleneck evidence, change, measured result.
 20. What makes a senior answer strong?
 - trade-offs + operational safety + measurable impact.

4) 20 Must-Know AWS Q and A

1. SQS vs Kafka?
 - queue simplicity vs log-streaming replay and partitioned scale.
2. RDS vs Aurora?
 - compatibility/control vs cloud-optimized managed performance and failover.
3. DynamoDB design first principle?
 - access patterns define keys and indexes.
4. ALB vs NLB?
 - L7 HTTP routing vs L4 high-performance TCP/UDP.

5. How to design secure service access?
 - IAM roles + private networking + least privilege.
6. How to reduce cloud cost quickly?
 - right-size compute, storage lifecycle, reserved/savings plans.
7. How to run safe deployment?
 - canary/blue-green with rollback signals.
8. How to handle multi-region DR?
 - explicit RTO/RPO + tested failover runbooks.
9. CloudWatch key alerts?
 - latency errors saturation and business KPI thresholds.
10. Why use DLQ?
 - isolate poison messages and preserve system flow.
11. How to absorb traffic spikes?
 - queue buffering and autoscaling.
12. How to avoid Lambda cold-start pain?
 - provisioned concurrency and optimized init.
13. Why private subnets for app tiers?
 - reduce attack surface and controlled egress.
14. How to secure secrets?
 - Secrets Manager/SSM + KMS + rotation.
15. Why use RDS Proxy?
 - connection management under burst traffic.
16. How to choose ECS vs EKS?
 - simplicity and ops overhead vs Kubernetes flexibility/ecosystem.
17. What causes queue backlog incidents?
 - producer-consumer mismatch, downstream slowness, retry storms.
18. How to reduce MTTR in AWS?
 - runbooks, dashboards, tracing, clear ownership.
19. How to present migration story?
 - risk plan, phased rollout, rollback, measurable outcome.
20. What do interviewers test with AWS questions?
 - architecture judgment and operational maturity.

5) Final Interview Delivery Tips

- Speak in compact structure: problem, approach, trade-off, outcome.
- Always include one metric when describing project impact.
- If unknown, state assumptions and proceed methodically.
- Keep answers practical, not purely theoretical.

6) Night-Before Checklist

- Prepare 8 stories with numbers.
- Revise top 30 rapid-fire questions.
- Sleep and hydration matter more than one extra hour of random reading.

7) L4 Rapid-Fire Addendum (Cross-Domain)

21. Difference between idempotency and deduplication?
 - Idempotency is operation safety on retries; dedup avoids repeated event processing.
22. Why can retry policies cause outages?
 - Coordinated retries amplify load and create retry storms.
23. What is the first thing to show in incident update?
 - Customer impact, scope, mitigation ETA, and owner.
24. When to choose eventual consistency?
 - User experience tolerant of short staleness and high write scale is needed.
25. How to discuss database migration risk?
 - Backfill strategy, dual-write/dual-read policy, cutover guardrails, rollback plan.
26. What is a mature timeout strategy?
 - Per-hop budget with total request deadline and cancellation propagation.
27. Why can autoscaling fail during spikes?
 - Bad metrics, cold starts, provisioning delay, downstream bottlenecks.
28. What is a strong architecture answer format?
 - Requirements -> constraints -> design -> trade-offs -> failure handling -> metrics.